

Explainer: Problem 1 (Rudin 7.6) — Alternating Series of Functions

What's going on?

We have the series $\sum_{n=1}^{\infty} (-1)^n \frac{x^2+n}{n^2}$. The signs alternate $(+, -, +, -, \dots)$, and the terms depend on x . We need to show two things:

1. On any bounded interval $[-M, M]$, the series converges **uniformly**.
2. For every single x , the series does **not** converge absolutely.

Key definitions

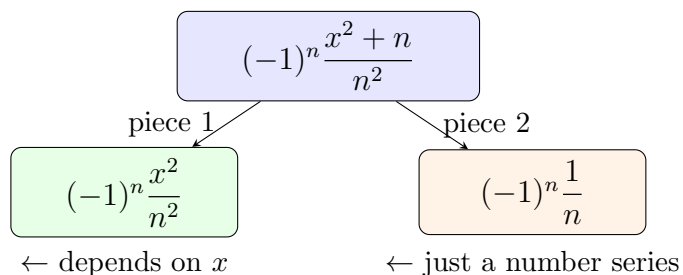
Uniform convergence of a series: $\sum f_n$ converges uniformly on E if the partial sums $S_N(x) = \sum_{n=1}^N f_n(x)$ converge uniformly, i.e. $\sup_{x \in E} |S(x) - S_N(x)| \rightarrow 0$.

Absolute convergence: $\sum f_n(x)$ converges absolutely at x if $\sum |f_n(x)| < \infty$.

Splitting the series

The key idea: split the n th term into two pieces:

$$(-1)^n \frac{x^2+n}{n^2} = (-1)^n \frac{x^2}{n^2} + (-1)^n \frac{1}{n}.$$



Strategy for uniform convergence

Piece 2 ($\sum(-1)^n/n$): This is just the alternating harmonic series — a fixed number, no x -dependence. It converges (to $-\ln 2$). A constant partial sum trivially converges uniformly.

Piece 1 ($\sum(-1)^n x^2/n^2$): Factor out x^2 to get $x^2 \sum(-1)^n/n^2$. On a bounded interval $|x| \leq M$, you can bound $|x^2/n^2| \leq M^2/n^2$. Since $\sum M^2/n^2 < \infty$, the **Weierstrass M -test** gives uniform (and absolute!) convergence.

Combining: a uniformly convergent series plus a (trivially) uniformly convergent constant gives uniform convergence.

Strategy for no absolute convergence

Look at $\left|(-1)^n \frac{x^2+n}{n^2}\right| = \frac{x^2+n}{n^2} = \frac{x^2}{n^2} + \frac{1}{n}$.

The x^2/n^2 part sums to something finite, but $\sum 1/n = \infty$ (harmonic series!). So $\sum |f_n(x)| = \infty$ for every x . The absolute series always diverges.

Takeaway

This is a nice example showing that a series can converge uniformly (thanks to cancellation from alternating signs) without converging absolutely. The alternating signs are doing real work here — they make $\sum(-1)^n/n$ converge even though $\sum 1/n$ doesn't.